Fiscal Year:	FY 2023	**Task Last Updated:**	FY 03/15/2023
PI Name:	Walker, Travis Ph.D.		
Project Title:	Investigation of Aspherical Magnetic Particles in Low-Gravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Rapid City	**State:**	SD
Zip Code:	57701-3901	**Congressional District:**	1
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017/2019 Physical Sciences NNH17ZTT001N-17PSI-F: Use of the NASA Physical Sciences Informatics System – Appendix F
Start Date:	12/22/2020	**End Date:**	12/21/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hatch, Tyler	**Contact Phone:**	216.433.5073
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/21/2023 per NSSC information (Ed., 1/12/23)		
Key Personnel Changes/Previous PI:	Per the Principal Investigator (PI) Mingyang Tan, Ph.D. is a CoInvestigator on this project. Dr. Tan is based at the University of Connecticut (Ed., 9/14/23).		
COI Name (Institution):	Tan, Mingyang (University of Connecticut)		
Grant/Contract No.:	80NSSC21K0293		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Magnetorheological fluids that are composed of suspensions of magnetic micro-/nano-particles are providing numerous opportunities for applications as smart fluids with properties that are tunable when subjected to external magnetic fields. However, investigations of these fluids are inhibited on Earth by the sedimentation of the particles that is caused by their inherent difference in density from the surrounding media. In this work, we use accelerated Stokesian dynamics simulations to investigate the dynamics and properties of coarsening of spherical and aspherical magnetic particles under different magnetic fields, and we compare our numerical results with experimental data of InSPACE, InSPACE-2, InSPACE-3, and InSPACE-3+.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Successful completion of this project could provide fundamental understanding of physics of magnetorheological (MR) fluids beyond the dynamics of coarsening of particles, e.g., the kinetics of phase transition, the transport properties of the system, the macroscopic magnetic properties of the overall system, and the microscopic magnetic properties of the local environment. Further, this work will not only predict the dynamics and rheological properties of MR fluids that were observed in InSPACE, InSPACE-2, InSPACE-3, and InSPACE-3+, but it will also provide insights into unexplored systems that are commonly seen in real applications. Beyond the fundamental knowledge of physics in MR fluids, advanced theoretical and numerical modeling and experimental validation of the behavior of MR fluids are also expected to provide robust metrics for the design of future MR systems. These systems could be applied in manufacturing of magnetic soft composites or in transporting therapeutic drug and gene delivery through tumor vasculature. Completion of this project can also be expected to direct future opportunities focusing on fundamental and applied aspects of making anisotropic composites, such as switchable anisotropic conductive barriers, magneto-optical devices, and liquid crystals, as well as opportunities to produce composite hydrogels using 3D additive manufacturing techniques. The Principal Investigator (PI) is currently modeling the microhydrodynamics of oblate spheroids in Newtonian fluids in collaboration with faculty in electrical engineering at Oregon State University (OSU) to engineer soft composites for high- frequency applications such as inductor and antennae. Anisotropic magnetic particles were chosen in this problem to enhance the magnetic properties, i.e., increased permeability of materials at high resonance frequency that allows for high-data-transmission rate. Completion of this work is projected to significantly impact efforts to perform voxel-by-voxel additive manufacturing via 3D printing of UV-curable magnetic inks. The PI is also studying a novel technique of micro-/nano-rheology, which measures magnetic susceptibility as a function of frequency of a magnetic colloidal suspension that is subjected to an oscillating magnetic field. The frequency-dependent magnetic susceptibility can be related to the rheological properties of the suspension. This work is expected to provide a novel method to measure the rheological properties of fluids that are difficult to measure in conventional macroscopic or microscopic rheological methods, e.g., in a limited amount of sample to collect for macroscopic rheological measurements or for an opaque sample that cannot be measured by conventional microscopic rheological methods that require visualization of the sample. The PI is funded by the NASA Established Program to Stimulate Competitive Research (EPSCoR) R3 to investigate microbial growth in rotating wall vessels by utilizing conventional Stokesian dynamics. Incorporation of accelerated Stokesian dynamics would advance the computational understanding of particle sedimentation, allowing detailed three-dimensional trajectories to be compared to fundamental theoretical calculations, which would verify our predictions of mass transfer during the introduction of an external stimulus. Completion of this work is projected to develop an in vivo technique for analysis of bio-fluids or tissues of patients by monitoring the change of rheological properties. The modeling and numerical methods that will be developed at the completion of this proposed work are expected to provide insights and directions to those problems that we are currently investigating, for example, the dependence of mechanical, diffusive, optical, and magnetic properties of composites on the choice of size, shape, and distribution of particles and the dependence of susceptibility of magnetic colloidal suspensions on the kinetics of the formation of microstructures.
Task Progress:	We developed an accelerated Stokesian dynamics (ASD) program that implements the spectral-accuracy method that calculates both the hydrodynamic and magnetic interactions accurately and efficiently. The rheological properties of a suspension of magnetic particles under constant magnetic fields are investigated. We have shown that the rheology of a suspension of magnetic particles depend on the value of Ma and the direction of the magnetic field. At low Mason number, when magnetic force dominates, the viscosity is enhanced when the field is in the gradient direction. Chains of particles are formed in the y direction, causing obstructions to the flow. However, when the field is in the x direction or z direction, chaining causes more free space for particles to flow, and thus reduced viscosities. Chains of particles formed in the x direction tend to slide by each other in the flow direction. Chains of particles formed in the z direction result in more void space in the x,y plane and therefore less obstruction to the rolling motion of the chains of particles. At high Mason number, when hydrodynamic force dominates, there are fewer chaining of particles. The configuration of particles is closer to a random one, no matter which direction the field points to, and the viscosity of three directions converge to the same value. That is to say, a suspension of magnetic particles shows shear thinning with an increase of Mason number when the field is in the gradient direction and shows shear thickening with an increase of Mason number when the field is in the flow direction or vorticity direction. The rheology of a suspension of magnetic particles depends on the value of Ma and the rotating field plane. Under a rotating magnetic field, the particles tend to form sheets of particles that align with the rotating plane of the field. When the field is rotating in the flow-gradient plane, at low Mason number (high magnetic field), the particles form sheets and rotate along this plane, which can be opposite to the direction of flow, hence the increased viscosity. As Ma increases, the magnetic field strength is reduced and the particles are less aligned to the plane of rotation, hence a reduced effective viscosity. This trend can be interpreted as shear-thinning, since viscosity decreases with increasing Mason number. When the field is rotating in the flow-vorticity plane, at low Mason number, the particles form sheets and rotate along the plane. These sheets are parallel to the flow direction, which are of less resistance to flow. As Mason number is increased, the particle sheets are not readily formed, hence more resistance to flow. This trend can be said to be shear-thickening as viscosity increases with increasing Mason number. When the field is rotating in the gradient-vorticity plane, at low Mason number, the particles form sheets and rotate along the plane. These sheets formed are perpendicular to the flow direction, which creates high resistance to flow. As Mason number is increased, the particle sheets are not readily formed, hence less resistance to flow. This trend can also be called shear-thinning, since

	viscosity decreases with increasing Mason number.
Bibliography Type:	Description: (Last Updated: 10/02/2023)
Abstracts for Journals and Proceedings	Tan M, Adeniran JA, Walker TW. "A computational investigation of the dynamics and rheological properties of MR fluids." 19th U.S. National Congress on Theoretical and Applied Mechanics, Austin, Texas, June 19-24, 2022. Abstracts. 19th U.S. National Congress on Theoretical and Applied Mechanics, Austin, Texas, June 19-24, 2022. , Jun-2022
Abstracts for Journals and Proceedings	Adeniran JA, Tan M, Walker TW. "Numerical simulations, scaling analysis, surrogate experiments, and analytic model of microbial growth in a rotating bioreactor." 96th American Chemical Society (ACS) Colloid and Surface Science Symposium, Golden, Colorado, July 10-13, 2022. Abstracts. 96th American Chemical Society (ACS) Colloid and Surface Science Symposium, Golden, Colorado, July 10-13, 2022. , Jul-2022
Abstracts for Journals and Proceedings	Tan M, Adeniran JA, Walker TW. "A computational investigation of the dynamics and rheological properties of MR fluids." 96th American Chemical Society (ACS) Colloid and Surface Science Symposium, Golden, Colorado, July 10-13, 2022. Abstracts. 96th American Chemical Society (ACS) Colloid and Surface Science Symposium, Golden, Colorado, July 10-13, 2022. , Jul-2022
Articles in Peer-reviewed Journals	Tan M, Adeniran JA, Walker TW. "Dynamics and rheological properties of suspensions of paramagnetic spherical particles under constant magnetic fields." Phys. Rev. Fluids. Accepted 2023 Mar 6. https://journals.aps.org/prfluids/accepted/ca079S83I9011c0472413a14d0b2cfdb5e474d714 , Mar-2023